

General form of the sine function: _____

We always plot 5 points!

- Amplitude
- Period
- Phase shift
- Vertical shift

Example 1: Graph one period of each of the following sine functions. Make sure to clearly label your axes. Show five points plotted and list all appropriate characteristics.

a. $f(x) = -3\sin x$

- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range

b. $f(x) = 10 \sin\left(\frac{x}{3}\right) - 1$

- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range

The period of a trig function is the time it takes to complete _____ cycle. The frequency is just the opposite; it's the number of complete cycles every 2π units. Because $y = \sin x$ and $y = \cos x$ cover exactly one complete cycle over this interval, their frequency is 1.

Period and frequency are inversely related. That is, the higher the frequency (more waves over units), the lower the period (shorter distance on the axis for each complete cycle).

$$\text{Period} = \frac{2\pi}{\text{Frequency}}$$

In our parent function, what represents the frequency?????

$$f(x) = a \sin(bx - c) + d \quad \text{or} \quad f(x) = a \sin\left(b\left(x - \frac{c}{b}\right)\right) + d$$

Example 2:

a. $f(x) = -2\sin\left(x + \frac{\pi}{2}\right)$

- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range
- Frequency

b. $f(x) = 2 \sin(3x - \pi) + 1$

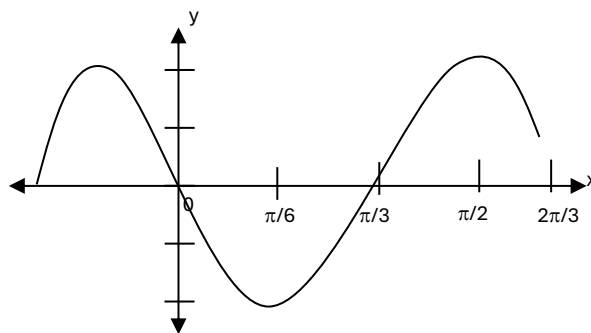
- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range
- Frequency

Example 3: Fill out the characteristics of the function. Then, write an equation based on the given starting points.

a. Starting at $x = \frac{\pi}{3}$

Amplitude _____ Period _____

Phase shift _____ Vertical shift _____



b. Starting at $x = \frac{\pi}{3}$

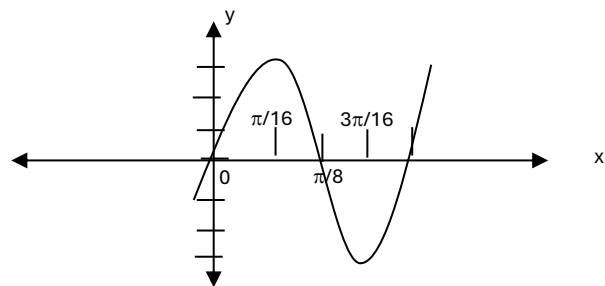
Amplitude _____ Period _____

Phase shift _____ Vertical shift _____

c. Fill out the characteristics. Then, write a sine equation to represent the function graphed.

Amplitude _____ Period _____

Phase shift _____ Vertical shift _____



AP Precalculus
Unit 2A Day 7 Practice

Graphing Sine Functions

1. Graph: $f(x) = 4 \sin\left(\frac{2}{3}x - \frac{\pi}{9}\right) + 2$

- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range

2. Graph: $f(x) = -2 + \frac{1}{2} \sin\left(\frac{\theta}{3} + \frac{2\pi}{3}\right)$

- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range

3. Graph: $f(x) = 3 \sin(4\theta) - 1$

- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range

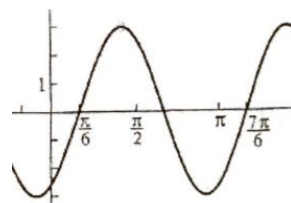
4. Graph $f(x) = 4 \sin(3x - 6\pi) + 1$

- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range

5. Fill out the characteristics. Then, write a sine equation to represent the functions graphed.

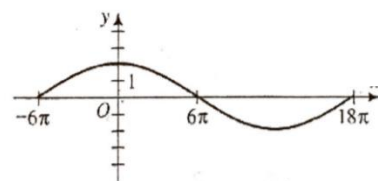
a. Amplitude _____ Period _____

Phase shift $\frac{\pi}{6}$ Vertical shift _____



c. Amplitude _____ Period _____

Phase shift 6π Vertical shift _____



6. Use the given characteristics to write an equation.

a. Sine curve with amplitude = $\frac{1}{2}$, period = $\frac{\pi}{2}$, phase shift = none, and vertical shift = up 7

b. Sine curve with amplitude = $\frac{2}{3}$, period = π , phase shift = $-\frac{\pi}{4}$, and vertical shift = none.

General form of the cosine function: _____

Just like graphing sine functions, we are going to use amplitude, period, phase shift, and vertical shift to graph transformed functions from the parent function of $y = \cos x$.

We are still going to graph five points, but cosine will start in a different place.... Where is that???

Example 1: Graph one period of each of the following sine functions. Make sure to clearly label your axes. Show five points plotted and list all appropriate characteristics.

a. $f(x) = 6\cos\left(\frac{1}{2}x - \pi\right)$

- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range
- Frequency

b. $y = 5\cos(2x + \pi) - 3$

- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range
- Frequency

c. $y = -\cos\left(\frac{x+\pi}{2}\right) + 4$

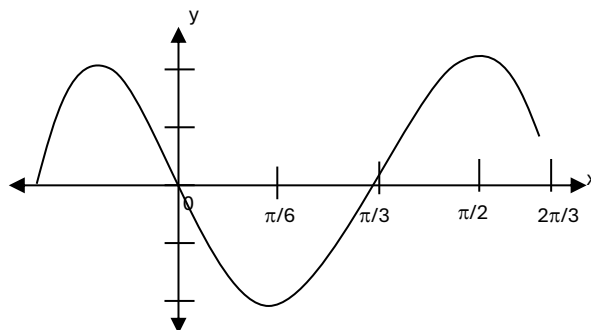
- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range
- Frequency

Example 2: Fill out the characteristics of the function. Then, write an equation based on the given starting points.

a. Starting at $x = \frac{\pi}{6}$

Amplitude _____ Period _____

Phase shift _____ Vertical shift _____



b. Starting at $x = \frac{\pi}{2}$

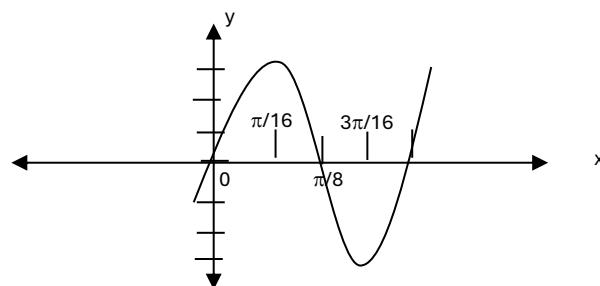
Amplitude _____ Period _____

Phase shift _____ Vertical shift _____

c. Fill out the characteristics. Then, write a cosine equation to represent the function graphed.

Amplitude _____ Period _____

Phase shift _____ Vertical shift _____



Find the amplitude, period, phase shift, vertical shift, domain, and range for each of the following functions.

1. $y = -5 \cos[2(x - \pi)] - 10$

2. $y = 4 \sin \frac{1}{3} \left(x - \frac{\pi}{4} \right)$

Graph one period of the following functions. Then, identify the domain, range, amplitude, period, phase shift, and vertical shift.

3. $y = \cos(3x + \pi)$

- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range
- Frequency

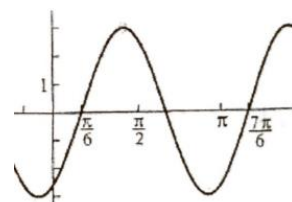
4. $y = 3 \cos(2x + 5\pi)$

- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range
- Frequency

5. Fill out the characteristics. Then, write a cosine equation to represent the functions graphed.

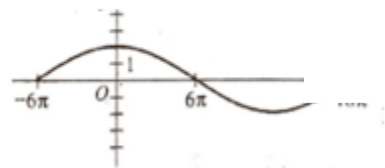
a. Amplitude _____ Period _____

Phase shift _____ Vertical shift _____



b. Amplitude _____ Period _____

Phase shift _____ Vertical shift _____



6. Use the given characteristics to write an equation.

a. Cosine curve with amplitude = $\frac{1}{2}$, period = $\frac{\pi}{2}$, phase shift = none, and vertical shift = up 7

b. Cosine curve with amplitude = $\frac{2}{3}$, period = π , phase shift = $-\frac{\pi}{4}$, and vertical shift = none.

Four Corners:

Analytical

Let $f(x) = 4 - 3 \cos(\pi x)$.

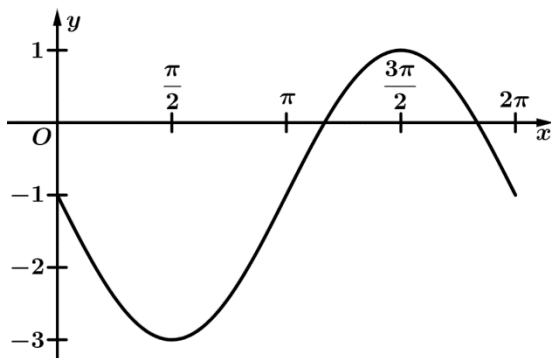
What is the maximum value of $f(x)$?

Numerical

The function g is defined by $g(x) = a \cos(2x) + d$, where a and d are constants. Selected values of g are shown in the table below. Find the missing values in the table.

x	0	$\pi/2$	$3\pi/4$	π	$3\pi/2$
$g(x)$	4		1		

Graphical



A portion of the graph of $h(x)$ is shown above, where $h(x) = a \sin(x) + d$. Find the values of the constants a and d .

Verbal

Let $k(x)$ be a sine function with period π , amplitude of 3, midline $y = -1$, and no other transformations of $y = \sin(x)$.

Write an equation for $k(x)$.

Big 10:

Complete the ten problems. Once you complete each problem, you will cross off one of the numbers in the box below. If done correctly, you will have all 10 numbers crossed off with no repeats.

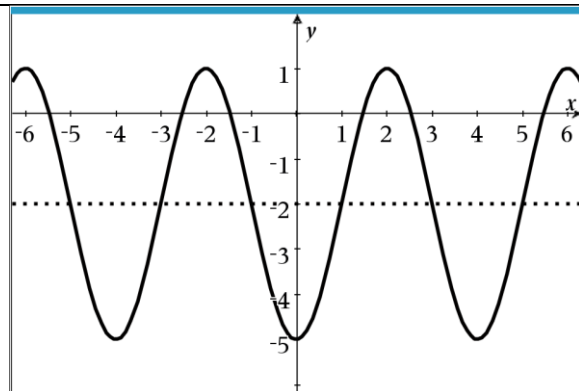
No Calculator

0	1	2	3	4	5	6	7	8	9
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Complete each problem.		Answer
A. The function $f(\theta) = \sin \theta$ is transformed to produce the graph of $g(\theta)$. What is the amplitude if $g(\theta) = -\sin(3\theta + \pi)$?	Amplitude is	
B. Find the period of the graph of the sinusoidal function $g(\theta) = 2\cos\left(\frac{\pi}{4}(\theta)\right) - 3$.	Period is	
C. The function $h(\theta) = \cos \theta$ is transformed to produce the graph of $j(\theta)$. What is the equation of the midline of the graph of the sinusoidal function $j(\theta) = 2\cos(4\theta - 3\pi) + 7$?	Midline is $y =$	
D. How many complete cycles does the sinusoid function $f(\theta) = 4\sin(9\theta) + 3$ have in the interval $[0, 2\pi]$?	Cycles =	
E. Find the frequency of the graph of the sinusoidal function $g(\theta) = 2\cos(10\pi(\theta)) - 3$.	Frequency is	

Use for F. and G.

In the figure to the right, the graph of a sinusoidal function of the form $g(\theta) = a \sin(b(\theta + c)) + d$, where a , b , c , and d are constants and $a > 0$. The graph of the midline, $y = -2$, is also shown as a dotted line.



F. What is the value of a ?	$a =$																			
G. What is the period of the sinusoidal function?	Period is																			
H. The function $h(\theta)=\cos \theta$ is transformed to produce the graph of $j(\theta)$. What is the horizontal translation (or phase shift) of the graph $h(\theta)$ in the sinusoidal function $j(\theta)=3h(2\pi\theta+4\pi)+7$?	Circle one: Right Left																			
I. Selected values of a sinusoidal function of the form $g(\theta)=a\sin(b(\theta+c))+d$ are given in the table to the right. It is known that a , b , c , and d are constants and $a\neq 0$. What is the value of d ?	<table><tr><td>θ</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>$f(\theta)$</td><td>9</td><td>6</td><td>3</td><td>6</td><td>9</td><td>6</td><td>3</td><td>6</td></tr></table>	θ	-1	0	1	2	3	4	5	6	$f(\theta)$	9	6	3	6	9	6	3	6	$d =$
θ	-1	0	1	2	3	4	5	6												
$f(\theta)$	9	6	3	6	9	6	3	6												
J. The graph of a sinusoidal function has the form of $k(\theta)=a\cos(b(\theta+c))+d$. If $k(\theta)=\frac{5}{\pi}\cos\left(\frac{\pi}{5}\theta-1\right)+8$, find a and c .	$a + c =$																			

Sinusoidal Function Equations Practice:

A. Answer the questions about each sinusoidal function.

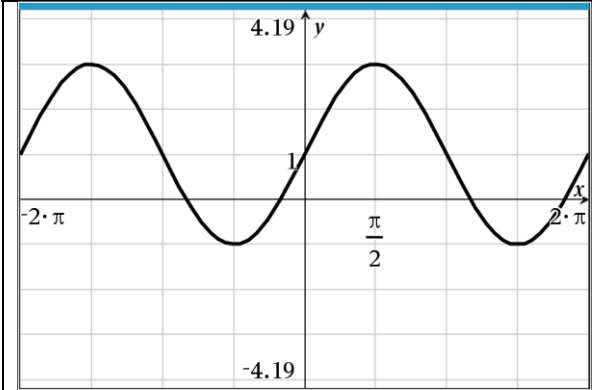
1. Transformations in the following order are performed on the graph of $f(\theta) = \cos \theta$ to produce the graph of $g(\theta)$. Write a function expression for $g(\theta)$.

A horizontal translation (phase shift) of $-\frac{\pi}{8}$ units, a horizontal dilation by a factor of 2 (period of 4π), a vertical dilation by a factor of 5 (amplitude of 5), and a vertical translation of 6 units.

θ	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{2}$
$f(\theta)$	1	-2	$4 - 3\sqrt{3}$	10

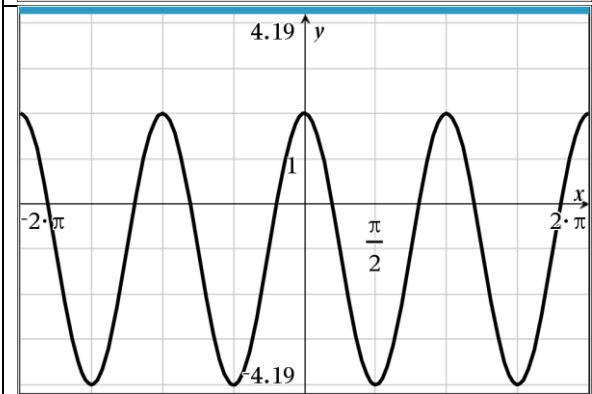
2. The function f can be modeled by a sine function of the form $f(\theta) = a \sin(\theta) + d$. Selected values of $f(\theta)$ are shown in the table above. What are the values of a and d for the function f ?

B. Write sine and cosine functions for the graphs shown.



1. The graph of a transformed **sine** function is shown to the right. The midline is $y = 1$. Write a possible function f for the graph shown.

2. Write the function in terms of cosine.



1. The graph of a transformed **cosine** function is shown to the right. The midline is $y = -1$. Write a possible function f for the graph shown.

2. Write the function in terms of sine.

Unit 2A Day 10 Notes

Today we will model periodic motion with trig functions 😊

Example 1: A Ferris wheel is 160 meters in diameter and boarded at its lowest point (6 O'Clock) from a platform which is 8 meters above ground. The wheel makes one full rotation every 16 minutes, and at time $t = 0$, when the wheel starts spinning, you are at the loading platform.

a. What is the period and what does it represent in this situation?

b. What is the equation of the midline?

c. What is the amplitude and what does it represent in this situation?

d. Write an equation to model the Ferris wheel's height above the ground as a function of time.

e. Use the information above to graph one period of the sinusoidal function.

Example 2: Over the past several years, the owner of a boutique on Aspen Avenue has observed a pattern in the amount of revenue for the store. The revenue reaches a maximum of about \$41,000 in April and a minimum of about \$26,000 in October. Suppose the months of the year are numbered 1 through 12.

a. What is the period and what does it represent in this situation?

b. What is the range and what does it represent in this situation?

c. Write a function that models the boutique's revenue during the year, where x corresponds to the month.

d. Use the information above to graph one period of the sinusoidal function.

You try! 😊 A buoy floating in the sea is bobbing in simple harmonic motion with period of 2 seconds and amplitude of 9 inches. Its displacement d from sea level at time $t=0$ seconds is -9 inches, and initially it moves upward (i.e. in the positive direction). The buoy bounces back and forth from a negative displacement to a positive displacement. Graph one period of the function. Then give the equation modeling the displacement d as a function of time t .

For each word problem, first sketch a graph to represent the periodic motion. Then, fill out the appropriate characteristics and write a function to represent the graph.

1. As you ride a Ferris wheel, your distance from the ground varies sinusoidally with time. When the last seat is filled and the Ferris wheel starts, you find that it will take you 6 seconds to reach the bottom, 3 feet above the ground. The wheel makes one revolution every 18 seconds, and the diameter of the wheel is 44 feet.

Period _____

Sketch:

Amplitude _____

Vertical Shift _____

Phase Shift _____

Equation:

2. Researchers found a creature from an alien planet. They noticed that its body temperature varied sinusoidally with time. Thirty-five minutes after they start timing, it reaches a high of 120°F . Twenty minutes after that it reaches its next low, 104°F .

Period _____

Sketch:

Amplitude _____

Vertical Shift _____

Phase Shift _____

Equation:

3. Mark Twain sat on the back of the Mississippi Queen and watched the paddlewheel. As the paddlewheel turned, he noticed that a point on the paddle blade moves in such a way that its distance, d , from the water's surface is a sinusoidal function of time. When his stopwatch read 5 seconds, the point was at its highest, 12 feet above the water's surface. The wheel's diameter was 14 feet and it completed a revolution every 16 seconds.

Period _____

Sketch:

Amplitude _____

Vertical Shift _____

Phase Shift _____

Equation:

4. As the motor on a jack turns, the walking beam rocks back and forth, pulling the pump rod in and out of a well. The pump is started at time $t = 0$ seconds. One second later, the rod is at its highest point 14 feet above the ground. It is at its next low point of 10 feet 2.5 seconds after that.

Period _____

Sketch:

Amplitude _____

Vertical Shift _____

Phase Shift _____

Equation:

Enhanced AA & AP
Unit 4A Test Review

Graph one period of each function and complete the information for each equation.

1. $y = \cos\left(3x - \frac{\pi}{2}\right)$ amp. = _____ period = _____ phase shift: _____ vertical shift: _____

2. $y = 3 \sin\left(x - \frac{\pi}{3}\right) - 2$ amp = _____ period = _____ phase shift: _____ vertical shift: _____

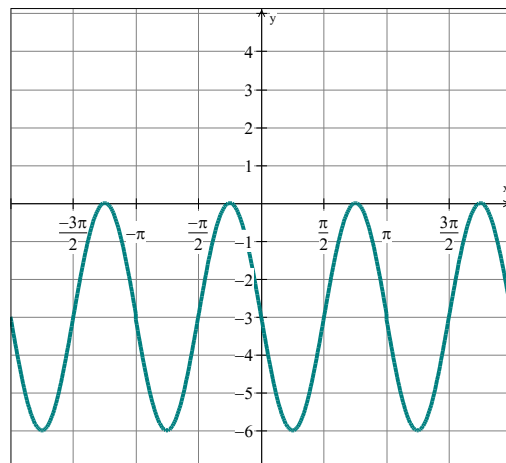
3. $y = -4 \cos\left(\frac{\theta}{2} + \pi\right) - 6$ a = _____ period = _____ phase shift: _____ vertical shift: _____

4. Find $\cos\theta$ the given that $\cot\theta = -\frac{6}{5}$ and $\sec < 0$.

5. Write an equation for each of the curves using the graph at the right.

Sine _____

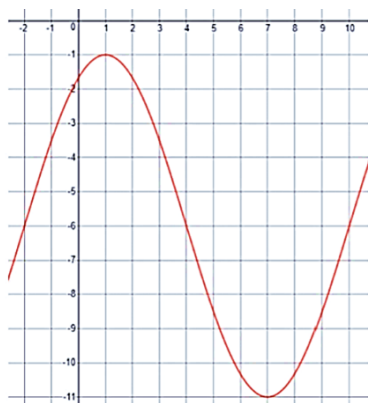
Cosine _____



6. Write a sine equation using the table

x	-5π	-3π	$-\pi$	π
y	2	-1	2	5

7. Write a cosine equation using the graph



Write a sine or cosine equation for the problems below.

8. One particular July 4th in Galveston, TX, high tide occurred at 9:36 a.m. At that time the water at the end of the 61st Street Pier was 2.7 meters deep. Low tide occurred at 3:48 p.m., at which time the water was only 2.1 meters deep. Assume that the depth of the water is a sinusoidal function of time with a period of half a lunar day (about 12 hours 24 minutes).

9. An earthquake occurred at 9:40 a.m. on Nov. 1, 1755, at Lisbon, Portugal, and started a tsunami in the ocean. It produced waves that traveled more than 540 ft/sec (370 mph) and reached a maximum height of 60 ft. The period of the waves was 30 minutes. Write an equation for the height of the wave as a function of time. Assume midnight is $t=0$.

10. Write an equation of a sine curve with amplitude = 3, period = 5, phase shift = $\frac{\pi}{4}$, vertical shift = up 2

11. Write an equation of a cosine curve with amplitude = $\frac{1}{4}$, period = $\frac{\pi}{2}$, phase shift = $\frac{\pi}{2}$, vertical shift = 0. Also, describe the transformations from the parent function $f(x) = \sin(x)$.

12. The minute hand of a clock is 6 inches long. How far does the tip of the minute hand move in 25 minutes. Leave answer in exact form.

13. A water sprinkler sprays water over a distance of 30 feet while rotating through an angle of 135° . What area of the lawn receives water? Leave in exact form.

14. Evaluate without a calculator.

$$\cos\left(\frac{2\pi}{3}\right)$$

$$\sec\left(-\frac{5\pi}{4}\right)$$

$$\sin\left(\frac{15\pi}{2}\right)$$

$$\cot\left(\frac{7\pi}{6}\right)$$

$$\csc\left(\frac{\pi}{2}\right)$$

$$\cos\left(-\frac{5\pi}{3}\right)$$

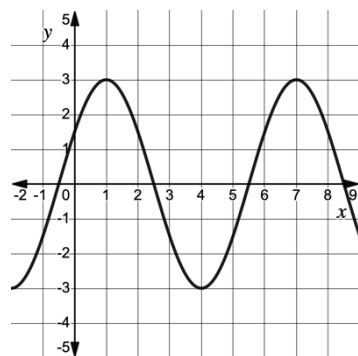
$$\sin\left(\frac{3\pi}{4}\right)$$

$$\tan 7\pi$$

$$\tan\left(-\frac{4\pi}{3}\right)$$

15. A portion of the graph of a periodic function $y = f(x)$ is shown. On each of the following intervals, determine if f is increasing or decreasing and if the graph of f is concave up or concave down.

Interval	Increasing or Decreasing?	Concave up or Concave down?
$22 < x < 23.5$		
$13 < x < 14.5$		
$38.5 < x < 40$		



16. The average monthly temperature in Denver, Colorado is recorded in the table below:

a. Using your calculator, find the sinusoidal regression function, $T(m)$. You will use the SinReg option. You will not need to change the default settings – Iterations: 3, Period: [Leave blank]. Round to 100th.

Month (m)	Temperature (Degrees Fahrenheit)
January	29.7
February	33.4
March	39
April	48.2
May	57.2
June	66.9
July	73.5
August	71.4
September	62.3
October	51.4
November	39
December	31

b. What is the midline of your function? What does this value represent in context of the data?

c. Find the period of your function? Does this number make sense in context of the data? Explain.

d. Use your Regression function to estimate the Temperature in Denver for May 6th of next year.